R&D or R vs. D?
Firm Innovation Strategy and Equity Ownership

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Abstract

We analyze a unique dataset that separately reports research and development expenditures for a large panel of public and private firms. Definitions of "research" and "development" in this dataset, respectively, correspond to definitions of knowledge "exploration" and "exploitation" in the innovation theory literature. We can thus test theories of how equity ownership status relates to innovation strategy. We find that public firms have greater research intensity than private firms, inconsistent with theories asserting private ownership is more conducive to exploration. We also find public firms invest more intensely in innovation of all sorts. These results suggest relaxed financing constraints enjoyed by public firms, as well as their diversified shareholder bases, make them more conducive to investing in all types of innovation. Reconciling several seemingly conflicting results in prior research, we find private-equity-owned firms, though not less innovative overall than other private firms, skew their innovation strategies toward development and away from research.

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1. Introduction

While there is a vast amount of empirical literature on firm investment in innovation, due to data limitations, said literature mostly focuses on firms' combined investments in both research *and* development. The question of how different types of firms vary in their focuses on research *versus* development has thus gone unexplored. We fill this gap in the literature by using restricted microdata from a survey conducted by the U.S. Census Bureau and the National Science Foundation (NSF) to study how the degree to which a firm engages in research *versus* development is related to its equity ownership type (e.g. public or private). Following the NSF (2015), we define research as, "The planned, systematic pursuit of new knowledge or understanding," and development as, "The systematic use of research and practical experience to produce new or significantly improved goods, services or processes." Since these definitions broadly correspond to the innovation literature's definition of "knowledge exploitation" or "exploitative innovation" (e.g., Chen, Gao, Hsu, and Li, 2019; Ferreira, Manso and Silva, 2012; Gao, Hsu and Li, 2018; Manso, 2011; and March, 1991), we use these data to test various theories of exploratory versus exploitative innovation and their relation to firm ownership status. In the process, we are able to reconcile several seemingly conflicting empirical findings in prior research on the relation between private equity fund ownership and innovative outcomes.[1]

We categorize a firm's equity ownership type in one of three ways: public (for publicly listed firms), private equity held (for firms owned by private equity buyout funds structured as limited partnerships and sponsored by specialized firms such as Bain Capital), or other private. We then examine how ownership type is related to firm exploration and exploitation of knowledge. We also examine differences in the propensity of firms with different types of equity ownership to apply for patents, have patents issued, introduce new products, and to introduce new production and logistics

[1] For example, there is mixed evidence regarding private equity short termism. Lerner, Sorensen, and Stromberg (2011) find that leveraged buyout firms do not reduce patenting activity; however, Yang and Zhang (2018) document evidence that leveraged buyout firms reduce new product announcements.

processes, as well as support activities. Finally, we use the survey data to explore how firms with different ownership types differ in how their research and development investments translate into new patents, products or processes.

We find that *public* firms tend to invest *more* in *research* than do private firms of similar size in the same industry. Since research investment roughly measures investment in "exploratory innovation," our findings are inconsistent with the theory of Ferreira et al. (2014), which predicts *private* firms are more likely to engage in exploration than are public firms. Moreover, we find that public firms, in addition to investing more in research, also invest more in development, as well as capital equipment and structures devoted to innovation (e.g., laboratory space and equipment). These findings suggest that public status is more conducive to investment in innovation overall. This is consistent with the notion that the alleviation of financing constraints that comes with public status facilitates innovation. Further, diversified shareholders are less averse to the greater idiosyncratic risk typical of investments in innovation. Accordingly, these findings are consistent with the notion that having a more diversified shareholder base, another characteristic of public firms, is more conducive to investment in innovation. We also find that public firms are more likely to introduce new processes, suggesting public status is more conducive to non-patentable innovation than is private status.

We further find that private equity-owned-firms tend to conduct just as much development as public firms, but significantly less research. Furthermore, though private equity-owned firms tend to invest about as much in overall innovation as do other types of private firms, private-equity-owned firms skew their innovation investments more toward development. This last result suggests that, while the relatively short holding period of private equity does not discourage innovation overall, consistent with the findings of Lerner, Sorensen, and Stromberg (2011), private equity ownership status is more conducive to innovation that exploits rather than generates knowledge. Finally, we fail to find any

evidence that proprietary disclosure cost considerations, related to innovation, influence the selection of firms into different types of equity ownership.

Finance and economic theory provide good *a priori* reasons to believe that firm ownership type could influence innovation strategy and *vice versa*. Ferreira, Manso and Silva (2014) construct a model in which public firms tend to engage in innovation that exploits existing ideas; whereas, private firms are more likely to explore new ideas. The model delivers this result because it assumes managers of public firms have the ability to cash out of some of their equity positions early; whereas, private firm managers do not. In addition, the model assumes exploitative innovation takes less time to generate payouts than does exploratory innovation. Given that exploratory and exploitative innovation are, respectively, analogous to research and development as defined by the NSF, this model predicts that private firms should engage in more research, and public firms should engage in more development. Gao, Hsu and Li's (2018) and Chen, Gao, Hsu, and Li (2019) analyses of patent data are consistent with this prediction. However, we do not find similar support for this prediction in our analysis.

There are also good *a priori* reasons to believe public firms may engage in more of both research and development relative to private firms. Since public firms are generally less financially constrained than private firms, it is plausible that it is less costly for them to finance investments in innovation. This, in turn, makes it more likely that they will invest in innovation of all sorts. In addition, it is likely that the shareholders of public firms are better diversified than that those of private firms, making the former more tolerant of idiosyncratic risk. Since it is intuitive that investments in exploratory innovation carry more idiosyncratic risk, the greater risk tolerance of public shareholders suggests public firms should conduct more research than private firms. Furthermore, it is not obvious that managerial compensation contracts at public firms cannot be written to incentivize innovation projects with more uncertainty and longer payout horizons, thereby mitigating the incentive effects of public ownership postulated to reduce exploratory innovation in Ferreira et al. (2012). Manso (2011) shows theoretically that such

innovation can be effectively incentivized so long as the manager's vesting period is sufficiently long and the payout function is sufficiently convex. Given that public firm compensation contracts empirically exhibit substantial variation along these characteristics (e.g., Gopalan, Milbourn, Song and Thakor, 2014), it is plausible that public firms can structure contracts to strongly incentivize research and exploratory innovation when it is optimal for them to do so. Our findings that public firms engage in both more research and more development than do private firms suggests they are, in fact, able to structure contracts to maintain sufficiently strong incentives for research such that the positive effect of their cheaper access to financing dominates whatever the negative effect of their public status may be.

At first glance, our results appear to be at odds with Gao, Hsu and LI (2018) and Chen, Gao, Hsu, and Li (2019), who find public status is associated with less novel patents, from which they conclude that public firms engage in less exploration. We note, however, that not all innovation is necessarily related to patenting. Because a patent application requires public disclosure of the technical details of the innovation, and patents expire, it is sometimes optimal to guard innovation as a trade secret rather than to patent it. Furthermore, it is generally less likely that a process can be patented than can a new product or product feature. Therefore, differences in the novelty between two firms' patents do not necessarily reflect a difference in the novelty of the all of their innovation activities. Our finding that public firms are more likely to develop new production and logistics processes than are private firms implies that studies comparing the innovation of public and private firms solely by comparing patent data are likely to be underestimating the relative novelty of public firm innovation.

Theory also suggests that private equity ownership is plausibly related to the amount and type of innovation activity, though the sign of that relation is not clear *a priori.* When private equity funds invest in a firm, they usually take a controlling interest and monitor the management closely. Hence, agency problems are muted (e.g., Jensen, 1989). Furthermore, since the firm remains private, the ability of their managers to cash out is limited. Based on these characteristics, the Ferreira et al. (2014) model

would suggest private equity ownership is highly conducive to exploratory innovation and research. On the other hand, private equity fund holding periods are relatively short (five years or less), leading some to allege they prioritize short-term-oriented activities (e.g., Kolasinski and Harford, 2014). Additionally, Yang and Zhang (2019) find significantly fewer new product announcements for private equity-held firms. Therefore, it is also plausible that private-equity-backed firms will prioritize development and exploitative innovation over research and exploration. Yet, Lerner, Sorensen and Stromberg (2011) find that firms increase patenting activity after joining a private equity fund portfolio, casting doubt on the proposition that private equity funds are short-term focused.

We find evidence reconciling both sides of this literature. Our results indicate that private equity-backed firms tend to skew their R&D spending toward development relative to other firm, both public and private, consistent with the hypothesis that they engage in more exploitation of knowledge than exploration. We also find that, in addition to investing less in research, private equity-backed firms are less likely to report introducing new products and new support services, consistent with Yang and Zhang (2019). On the other hand, we find that they apply for more patents, consistent with Lerner et al. (2011). Taken together, these results suggest that while private equity-backed firms are not less innovative than others, and in fact, measured using only patent data, are more innovative, their innovation skews toward marginal product improvements through knowledge exploitation.

Another plausible hypothesis is that firms utilize private equity to finance innovation for which disclosure costs are particularly high. Firms introducing new products to markets have an incentive to hide information related to their operating performance in order to not attract rival entrants into their product space. We therefore hypothesize that firms with high investment needs, and who are highly active in introducing new products to the market, as well as firms with high investment needs and high reliance on trade secrets, are more likely to utilize private equity financing. We fail to find evidence supporting the abovementioned hypothesis.

We next examine how past and contemporaneous investments in R&D relate to various measures of innovation output: Patent filings and issuances and new product and process introductions. We regress each innovation-type output on contemporaneous research and development, as well as three lags. Consistent with the NSF definitions of research and development, we find that development has a stronger association with all innovation outputs than research. Curiously, however, we find that it is the contemporaneous value that has the strongest effect.[2] Finally, we find that the association between patent issuances and applications and contemporaneous research and development is strongest for public firms. We conclude that public status is positively associated with higher levels of innovation investment quality as well as quantity.

The associations we uncover between innovation strategy and ownership type could reflect a causal effect of ownership type, or could reflect firm selection into the ownership type that is optimal for its innovation strategy. To help distinguish between selection and causation, we examine whether innovation strategy changes when firms change their ownership status. That is, we examine how research, development and the propensity to introduce new products and processes changes as firms change status in the following ways: Public to private-equity-backed, private-equity-backed to public, other private to public, and finally, private-equity-backed to other private. While we find some evidence that firms moving from public to private-equity-backed status do increase development expenses, we fail to find strong evidence that changes in status are related to changes in innovation strategy in the short term otherwise. We thus interpret our main results as evidence that firms tend to select into the ownership type that is optimal for their innovation strategies, rather than evidence that ownership type has a causal effect on innovation strategy. Analyses on how the innovation strategy of private equity-

[2] Notably, we find a similar pattern with research. That is, even though research overall has a weaker effect on innovation outcomes than development, the contemporaneous value has a stronger effect than the lags.

owned firms is related to their new ownership status after they are sold by their private equity owners further supports the selection hypothesis.

2. Data

Our sample begins with all observations within the Business, Research, Development and Innovation Survey (BRDIS) database from 2009-2015, from the U.S. Census Bureau. The survey selects a sample of public and private firms, though it oversamples larger firms and firms with positive research and development expenses. Approximately one-third of Compustat firms are included in the survey. Certain firms are also sampled with certainty in any given year. Details of the sample selection procedure can be found at the NSF website.[3] Responding to the survey is required by law, so selection bias due to failure to respond is not a concern. Nevertheless, the Census and NSF adjust their data for non-response bias, and we use the adjusted data in all of our tests.

We merge the BRDIS survey with the Census' Longitudinal Business Database (or LBD), using internal Census firm identifiers, in order to obtain payroll and employment for each firm-year observation. Since the LBD reports data at the establishment level, for firms with multiple establishments, we sum payroll and employment to obtain a firm level number. We exclude all firm-year observations with zero payroll from our sample. Since the survey includes a large number of extremely small private firms that are not comparable to public firms or private-equity-owned firms, we drop from the sample any firm whose payroll is smaller than the first percentile payroll of BRDIS firms that we identify as belonging to a private equity portfolio. (We describe this identification process in depth below.) Since both payroll and employment are always positive in our sample, and yet highly skewed, we use the natural logarithmic transformations of these variables to control for firm size in our regressions, along with log sales from the BRDIS. We also utilize the age of each firm from the LBD

[3] https://www.nsf.gov/statistics/srvyindustry/#qs&sd

(measured as the age of its oldest establishment) and use its natural logarithmic transformation, and its square, as controls. (We use the square to control for the possibility of non-linear firm lifecycle effects.)

From the 2009-2015 BRDIS survey data files, we obtain, for each firm-year firm reported, domestic research, domestic development and worldwide combined R&D. Unfortunately, the survey does not separately collect information on worldwide research and worldwide development. We thus scale domestic research and domestic development by the ratio of worldwide-to-domestic, combined R&D, obtaining separate proxies for worldwide research and worldwide development. For over 95% of the observations in our sample, however, R&D is domestic. If a firm has missing worldwide combined R&D, we assume research and development are both zero. If a firm reports non-zero worldwide R&D, but the values of domestic research or development are missing, we drop the observation. We also obtain the firm's reported domestic expenditures on capital equipment and structures devoted to research or development and label it R&D CAPX. If the value is missing, we assume it is zero. If it is not missing, we scale it by the ratio of worldwide-to-domestic, combined R&D to obtain a proxy for worldwide R&D CAPX. Since the distributions of research, development and R&D CAPX are highly skewed, but also often take the value of zero, we use the inverse hyperbolic sine (asinh) transformation of all these variables in all of our tests. Coefficients from regressions using asinh transformed data have a similar interpretation to coefficients estimated on log-transformed data. However, asinh has the advantage of being defined at zero. In robustness tests, we find our estimates are nearly identical, both in magnitude and statistical significance when we use log transformations. Where variables could potentially be zero, a one is added and then the logarithm is taken.

Total worldwide sales, the number of patent applications, the number of patents issued in the year, and the number of scientists, engineers and technicians employed by the firm for R&D activities are also obtained from BRDIS. These variables are also highly skewed, so as in the case of sales, we take the natural logarithm. For the patent and personnel counts, we add one and take the natural logarithm,

as is standard for count data that can take the value of zero (unlike for continuous data, for count data the so-called "started log" transformation, wherein one is added before taking logarithm, has a straightforward interpretation). We also construct dummy variables indicating whether the number of patents issued or applied for is positive, as well as dummy variables for observations where patent count data are missing. If the firm has missing data on patent or personnel count, we set it to zero.

We also create dummy variables related to other types of innovation outputs. That is, we define dummy variables to indicate whether the firm reported introducing new products or services, new production processes, new logistics processes, or new support activities. We also define dummy variables indicating whether the firm indicates utility patents, design patents and trade secrets are somewhat important or very important to its business. All these dummy variables take the value of one if the answer is in the affirmative, and are zero otherwise, even if data are missing. For the trade secret and patent importance dummies, we also create dummies where responses are missing.

We use the Preqin database, a global directory of private equity portfolio companies, to identify all firms within BRDIS from 2009-2015 that are part of private equity buyout fund portfolios. To that end, we first merge BRDIS with the LBD and Standard Statistical Establishment List/Business Register (SSEL) in order to retrieve the names and addresses of each BRDIS firm establishment. Then, for each year, we use text matching algorithms to compute a similarity score between the name of every BRDIS firm establishment and every firm in Preqin that is flagged as being part of a private equity portfolio within that year. We strip all names of generic words common to company names, such as "Corp," "Corporation," "Company," "Group," "LLC," and "Industries," before computing the similarity score. We then manually check the names for every potential match with a similarity score of 0.5 or greater (on a scale of 0-1) and where the Preqin industry matches the NAICS code in the Census datasets. We also consider the state, zip code and street addresses in making this match. If the state and zip code match, and the names are sufficiently close so that the firms are highly likely, in our judgement, to be the same,

we manually designate the pair to be a true match. Because firms sometimes relocate, we also identify the pair as a true match if the names and industries are a near perfect match, even when the addresses do not match. If the states and zip codes match exactly, we manually examine each potential matched pair even if the similarity score is as low as 0.3, and designate the pair as a true match if the names and industries are sufficiently close in our judgement.

Once we identify the BRDIS firms that are in private equity portfolios in a given year from 2009-2015, we use the Compustat–SSEL Bridge file to identify firms that are publicly listed in a given year. Since the bridge file only extends to 2011, we extend it ourselves to 2015. We do this by identifying the firms deleted from Compustat in 2012-2015 and removing them from the bridge file in the appropriate year. We then examine which firms were added to Compustat in each of those years and match them with firms in the LBD based on employer identification number (EIN). Those that we cannot match on EIN we match using historical names, addresses and NAICS codes from Compustat and the SSEL. Once we match these firms, we add them to the Compustat–SSEL Bridge file in the appropriate year.

Once we obtain the ownership status of every firm in the BRDIS database, we remove "other private" firms that have sales below the first percentile of those in the "private equity-owned" category. This ensures that there is overlap in firm size between the types, as the very small firms in the "other private" category are likely to be fundamentally different from the others. In Table 1, we present descriptive statistics on all data used in our tests, delineated by the three ownership types: public, private equity owned and other private.

3. Methods and Results

We first examine how various innovation variables are related to the ownership status of each firm in the BRDIS database. To that end, we run linear probability models of the form:

$$Ownership_{i,t} = \alpha + \beta_1 \operatorname{asinh}(resesarch_{i,t}) + \beta_2 \operatorname{asinh}(development_{i,t}) + \beta_3 \operatorname{asinh}(R\&DCAPX_{i,t}) +$$
$$\beta_4 Trade\ Secrets\ Important_{i,t} + \beta_5 Design\ Patents\ Important_{i,t} +$$
$$\beta_6 Utility\ Patents\ Important_{i,t} + \beta_7 \ln(1 + \#Patents\ Filed_{i,t}) + \beta_8 I(\#Patents\ Filed_{i,t} > 0) +$$
$$\beta_9 \ln(1 + \#Patents\ Issued_{i,t}) + \beta_{10} I(\#Patents\ Issued_{i,t} > 0) + \beta_{11} New\ Products_{i,t} +$$
$$\beta_{12} NewProductionProcesses_{i,t} + \beta_{13} NewLogisticsProcesses_{i,t} + \beta_{14} NewSupportActivities_{i,t} +$$
$$\beta_{15} \ln(sales_{i,t}) + \beta_{16} \ln(payroll_{i,t}) + \beta_{17} \ln(employees_{i,t}) + \beta_{18} \ln(age_{i,t}) + \beta_{19} \ln(age_{i,t})^2 +$$
$$\mathbf{\Gamma' MissingVariableDummies} + \varepsilon_{i,t} \qquad (1)$$

Where *i* indexes the firm, *t* indexes the year. $Ownership_{i,t}$ is a categorical variable indicating either private equity, public or other ownership of the firm. $I()$ is an indicator function, taking the value of one if the criterion in the parentheses is satisfied and zero otherwise. We run three different versions of this model. We run linear probability models because some of the dependent variables of ownership type are sparse in some specifications, and it is well known that sparse dependent variables can bias coefficients in categorical data specifications (such as logit and probit) that utilize a non-linear link function.

In the first, the dependent variable takes the value of one if the firm is public, and we estimate the model for our full sample. This model thus estimates how innovation strategy is related to public status versus both kinds of private statuses. In the second model, the dependent variable takes the value of one if the firm is held by a private equity buyout fund and is zero if the firm is public, and we estimate this model on the subsample of firm-year observations that are either public or held by private equity buyout funds. This model thus estimates how innovation strategy is related to private equity-held status versus public status. Finally, in the third model, we also use an indicator for private equity buyout fund held status as the dependent variable, but now we run it on the subsample of firm-year observations that only include private firms. This model thus estimates how innovation strategy is

related to being held by a private equity fund versus being held by another kind of private owner. In all cases, we conservatively cluster standard errors by four-digit NAICS codes, thereby making our inferences robust to arbitrary heteroscedasticity, serial correlation, and cross-sectional correlation within the broad industry group. We also include industry-by-year fixed effects, with industries defined by six-digit NAICS codes. The results are in Panel A of Table 2.

As can be seen in column 1 of Panel A of Table 2, public firms engage in more research relative to firms of similar size and in the same industry and year. (We control for three size measures and industry-by-year fixed effects.) This finding is inconsistent with Ferreira et al. (2014), whose theory has the empirical implication that public firms are less likely to engage in exploratory innovation. Another interesting finding is that public firms also engage in more development and invest more in capital goods and structures necessary to innovate, as indicated by the positive coefficients on development and R&D CAPX. Overall, then, we find that public firms invest more in innovation across the board, suggesting public status is more conducive to high innovation intensity. Plausible mechanisms for this finding include public firms having fewer financing constraints or their having a more risk-tolerant shareholder base due to greater diversification. We leave the task of pinning down the mechanism to future research.

Turning our attention to the second and third columns of Panel A of Table 2, we see that private equity-owned firms differ considerably from others in their focus on development. Note that in the specification comparing private equity-owned firms to public firms, there is no significant difference in development investment, but private equity-owned firms conduct significantly less research than public firms. Likewise, private equity-owned firms conduct more development and less research than other private firms, as indicated by parameter estimates in column 3 of Table 2.

A consistent picture of the three types of ownerships thus emerges. Public firms engage in more investment in all types of innovation relative to private firms. Private firms not owned by private equity buyout funds invest less in innovation than public firms, but conduct roughly equal proportions of exploratory and exploitative innovation. By contrast, private-equity-owned firms conduct high levels of exploitative innovation, comparable to that of public firms, but do little exploratory innovation. This finding is consistent with private-equity ownership being less conducive to an exploratory innovation strategy due to the shorter horizons of private equity investors. It is also consistent with prior literature finding private equity firms tend to focus on operating improvements.

Notice, in Table 2, public firms are less likely than private firms to report that utility patents are important for the business. We also document that they are less likely to have zero patents issued or to apply for zero patents in a given year. (Though this is conditional on having positive patents issued or applied for; they tend to have more.) Finally, we find public firms are more likely to introduce both new production and logistics processes, which may prove difficult-to-patent. Therefore, the results suggests that public firms are more likely to engage in difficult-to-patent innovation than private firms. Accordingly, prior research comparing public and private firm innovation via patenting activity are potentially missing important aspects of innovation. Results in columns 2 and 3 of Table 2 further suggest that private-equity-owned firms sit midway between public and other private firms in terms of their investments in difficult-to-patent innovation. That public firms are more willing to engage in difficult-to-patent innovation, which is thereby more vulnerable to rival imitation due to lack of intellectual property protection, is inconsistent with the hypothesis that disclosure costs associated with innovation are important determinants of public status.

We also run another set of three linear probability regressions similar to the above. However, we use the log count of scientists, technicians and engineers, instead of research and development, to

measure intensity of investment in innovation. The results, in Panel B of Table 2, are largely consistent with the results in Panel A of Table 2.

Another interesting question to investigate is whether different types of equity ownership are associated with divergent levels of efficiency in converting innovation input to output. To that end, we estimate specifications in which we regress innovation outputs on innovation inputs, as well as three lags of each, along with our size controls:

$$Y_{i,t} = a + \mathbf{B_1}'\mathbf{L}(0,3,\operatorname{asinh}(Research_i)) + \mathbf{B_1}'\mathbf{L}(0,3,\operatorname{asinh}(Development_i)) + \mathbf{\Gamma}'\mathbf{L}(1,3,Y_i) + \mathbf{Controls_{i,t}} + \varepsilon_{i,t} \qquad (2)$$

Where **L(***s,n,X***)** describes a vector that includes all lags of *X* between *s* and *n*, inclusive. A value of *s*=0 indicates the vector includes the contemporaneous value of *X.* Outputs *Y* include the started log of patents issued, the started log of patent applications and dummy variables indicating the introduction of new products, processes and support activities. In the case of the specification for patents issued, we also include three lags of the started log of patents applied. Controls are included for size, as well as dummies for public and private equity held statuses. Results are in Panel A of Table 3. The specification for patents issued also includes three lags of transformed patent applications, the parameters of which are suppressed for brevity.

As can be seen in Panel A of Table 3, there is a lot of persistence in innovation output, as lags of the dependent variable are generally significant in all cases. Markedly, only contemporaneous input variables (research and development) appear to matter for output, though the effect of development is stronger, as expected. Additionally, we find that private equity held firms apply for, and are issued, significantly more patents, consistent with Lerner, Sorensen, and Stromberg (2011). However, we also find that private equity-held firms introduce fewer new products and new support services, consistent with Yang and Zhang (2019).

Due to the findings in Panel A of Table 3, we only interact the contemporaneous value of the input variables with public and private equity held status in Panels B and C (for development and research, respectively). We only report the interaction term parameters for the sake of brevity, but the same independent variables reported in Panel A are included in these specifications. As can be seen, it appears that public firms are most efficient at turning both research and development spending into patents. Nonetheless, public firms appear to be less efficient at turning research and development into new products, suggesting their investment in innovation is more focused on processes and product improvements.

To get a better sense of whether any effects we find are due to firms selecting the ownership type most appropriate for their innovation strategies, or whether ownership type is causing the innovation strategy, we examine whether changes in ownership status correlate with changes in innovation strategy. To that end, we take the subsample of firms in our panel for which there are at least three consecutive observations for the firm. Then, for each observation, we take the difference between the variables in the year after and the year prior to the current year. We then regress these changes of innovation strategy on dummy variables indicating a change in ownership status, along with changes in our log-transformed size measures (payroll, sales and employment). The resulting specification is as follows:

$$\Delta Y_{i,(t+1)-(t-1)} =$$

$$\alpha_{k,t} + \beta_1 pub_to_pe_{i,t} + \beta_2 pe_to_pub_{i,t} + \beta_3 pe_to_othpriv_{i,t} + \beta_4 othpriv_to_pe_{i,t} +$$

$$\beta_5 othpriv_to_pub_{i,t} + \beta_6 \Delta \ln(sales)_{i,(t+1)-(t-1)} + \beta_7 \Delta \ln(payroll)_{i,(t+1)-(t-1)} + \varepsilon_{i,(t+1)-(t-1)} \quad (3)$$

Where *Y* takes on the following innovation variables, each in a separate specification: $\text{asinh}(research)$, $\text{asinh}(development)$, $\ln(patents\ aplied)$, $\ln(patents\ issued)$, *NewProducts, NewProductionProcesses, NewLogisticsProcesses,* and *NewSupportServices.* The Δ operator signifies the

change between the year after ($t+1$) and the year prior ($t-1$) to year t. The change in ownership status dummies are all contemporaneous (i.e. year t). Thus, the coefficients on the change in status dummies, represent the estimated effect of a change on the lead value of the dependent variable relative to the lag value. All coefficients are estimated using OLS with industry-by-year fixed effects, with industries defined by six-digit NAICS codes. Reported standard errors are clustered by four-digit NAICS codes to be conservative. Descriptive statistics on the smaller sample of firm-year observations, wherein these regressions are computed, are in Table 4. Results from the regressions are in Table 5.

To draw inferences from the specification given in equation 3, we make the identifying assumption that the firm's optimal innovation strategy is somewhat persistent, and does not change radically from year to year. We also assume that if there is a causal effect of ownership type on innovation strategy, it is likely to manifest itself significantly within a year. Thus, if our estimates of equation 1 uncover an association between ownership type and innovation strategy, but our estimates of equation 3 do not uncover an association between changes in year-by-year innovation strategy in response to changes in ownership type, we will conclude the effects found in equation 1 are the result of selection. If we do find effects in equation 2 in a similar direction to the effects found in equation 1, we will infer that ownership type likely has some causal effect, though we will not be able to precisely measure it.

As can be seen in Table 5, the coefficient on the *pub_to_pe* dummy is positive and significant, suggesting that development increases as firms move from public to private-equity-held status. In the new production process regressions, the coefficients on *othpriv_to_pub* and *pe_to_pub* are positive and significant, suggesting that firms become more likely to engage in process innovation as they transition from private to public status. Hence, it is possible that changes in public to private equity status result in more development spending; whereas, transitions into public status result in more process innovation. Both of the aforementioned results are consistent with our linear probability models, which suggest that

private equity-owned firms tend to skew their innovation investments toward development, and public firms tend to invest more in process innovation. However, most of the changes in status dummies are statistically insignificant. We thus infer that our findings in Table 2 more likely reflect firm selection into the type of ownership status that best suits its innovation strategy, rather than a causal effect of ownership status on innovation.

To further study whether innovative activities have a causal effect on the ownership type of a firm, we study the exit outcomes of private equity-held leveraged buyout investments. We test whether innovative activities are significant determinants of the exit of a private equity firm to a public operating company (trade sale), to public markets (IPO), to another private equity firm (secondary buyout), to a private operating firm (other private), or a or write-off or restructuring (bankrupt). We regress an exit-type indicator on various innovation and control variables in a system of jointly estimated linear probability models for deals exited between 2010 and 2016 that we were able to match to the BRDIS. We average all innovation and control variables over the period over which the firm was in the private equity portfolio, for all years for which we have data. We include 3-digit NAICS industry and investment year fixed effects, with standard errors clustered by 3-digit NAICS code.

As reported in Table 6, we find the innovation activities of a portfolio company significantly impact exit type. Specifically, we find that private equity-held firms that are smaller, have greater research intensity, are more likely to introduce new products, place low importance on utility patents, and place a higher importance on trade secrets are significantly more likely to be purchased in a public trade sale relative to other outcomes. These results are consistent with public operating companies acquiring firms that are potentially better fits into their established portfolio, as well as research being important to the business of large public operating companies. However, we also find that larger private equity-held firms (as measured by payroll) are more likely to go public through an IPO. However, we find that they are not more innovative, on average, than firms exited through other means. Finally,

and somewhat consistent with the findings of Lerner, Sorensen, and Stromberg (2011) we find that private equity backed firms that report utility patents are important to their business are more likely to be sold to other private equity firms in a secondary buyout. Additionally, and contrary to the disclosure cost hypothesis, we find that a secondary sale is less likely if trade secrets are important to the private equity-backed firm.

4. Conclusion

The literature on innovation theories has long distinguished between "exploration" and "exploitation" of knowledge. Due to the fact that firms generally report aggregated spending of their innovation activities, it has been difficult for the empirical literature to test these theories. Some prior empirical work has made some progress by examining the nature of patents filed for by firms, but many firm innovations, particularly process innovations, are difficult-to-patent. Patenting activity, therefore, does not reflect all innovation. Utilizing a unique dataset in which research and development are separately reported, we can distinctly observe firm investment in exploratory and exploitative innovation. Contrary to some prominent theories of innovation, we find public firms have greater investment intensity in exploratory innovation. We also find public firms tend to invest more in innovation of all types, suggesting that several unique features of public firm status make them more conducive to investment in innovation overall. Private equity-backed firms, consistent with shorter horizons due to limited holdings periods, tend to skew their innovation activities toward exploitation of knowledge rather than exploration. Finally, analyses of how changes in innovation strategy relate to changes in ownership status suggest our findings are due to firms selecting into the ownership status that best suits their innovation strategies, rather than a causal effect.

References

Chen, G., Gao, H., Hsu, P., Li, K., 2019. "Going Public and Innovation Strategy: The Role of Ownership Structure." INSEAD Working Paper.

Ferreira, D., Manso, G. and Silva, A.C., 2014. "Incentives to innovate and the decision to go public or private." Review of Financial Studies 21 (1), 256-300.

Gopalan, R., Milbourn, T., Song, F., Thakor, A.V., 2014. "The duration of executive compensation." The Journal of Finance, 69, 2777–2817.

Gao, H., Hsu, P., Li, K., 2018. "Innovation Strategy of Private Firms." Fudan University Working Paper.

Harford, J. and Kolasinski, A., 2014, "Do Private Equity Returns Result from Wealth Transfers and Short-termism? Evidence from a Comprehensive Sample of Large Buyouts and Exit Outcomes," Management Science, 60(4), 888-902.

Jensen, M., 1986. "Agency Costs of Free Cash Flow, Corporate Finance and Takeovers." American Economic Review, 76(2): 323–29.

Jensen, M., 1989. "Eclipse of the Public Corporation." Harvard Business Review, 67(5):61–74.

Lerner, J., Sorensen, M. and Stromberg, P., 2011. "Private Equity and Long-Run Investment: the Case of Innovation." The Journal of Finance, 66(2), 445-477.

Manso, G. 2011. Motivating Innovation. The Journal of Finance 66:1283–60.

March, J. G., 1991. Exploration and Exploitation in Organizational Learning. Organization Science 2, 71-78.

National Science Foundation. 2015. "2015 Business R&D and Innovation Survey." Retrieved from https://www.nsf.gov/statistics/srvyindustry/about/brdis/surveys/srvybrdis-2015-BRDI-1.pdf

Yang, W. and Zhang, F., 2019. "Private Equity Funds and Firm Products." University of South Carolina Working Paper.

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Table 1

Descriptive Statistics by Ownership Type. BRDIS Firm-Year Observations from 2009-2015.

Panel A: Public Firm-Year Observations 2009-2015. N = 11,000*.

	mean	std dev.	skewness	kurtosis
asinh(research)	3.948e+00	4.839e+00	5.750e-01	1.633e+00
asinh(development)	5.938e+00	5.519e+00	1.501e-02	1.291e+00
ln(1+Engineers)	3.681e-01	1.349e+00	3.963e+00	1.880e+01
ln(1+Scientists)	1.652e-01	8.299e-01	5.844e+00	3.992e+01
ln(1+Technicians)	1.731e+00	2.213e+00	9.436e-01	2.680e+00
asinh(R&DCAPX)	3.970e+00	4.487e+00	4.536e-01	1.568e+00
Trade Secret Dummy	2.222e-01	4.158e-01	1.336e+00	2.785e+00
Trade Secret Missing	4.019e-01	4.903e-01	4.000e-01	1.160e+00
Design Patents Important Dummy	4.802e-01	4.996e-01	7.911e-02	1.006e+00
Design Patents Important Missing	4.004e-01	4.900e-01	4.067e-01	1.165e+00
Utility Patents Important Dummy	2.545e-01	4.356e-01	1.127e+00	2.271e+00
Utility Patents Important Missing	3.994e-01	4.898e-01	4.106e-01	1.169e+00
ln(1+Patents Applied)	1.266e+00	1.869e+00	1.355e+00	3.883e+00
Patents Applied>0 Dummy	4.037e-01	4.907e-01	3.926e-01	1.154e+00
Missing Patent Applications	3.098e-01	4.624e-01	8.225e-01	1.676e+00
ln(1+Patents Issued)	1.085e+00	1.729e+00	1.591e+00	4.745e+00
Patents Issued > 0 Dummy	3.741e-01	4.839e-01	5.205e-01	1.271e+00
Patents Issued Missing	3.215e-01	4.671e-01	7.643e-01	1.584e+00
New Products Introduced Dummy	3.621e-01	4.806e-01	5.737e-01	1.329e+00
New Production Processes Dummy	2.057e-01	4.042e-01	1.456e+00	3.120e+00
New Logistics Processes Dummy	1.530e-01	3.600e-01	1.927e+00	4.715e+00
New Support Activities Dummy	2.071e-01	4.052e-01	1.446e+00	3.090e+00
ln(sales)	1.379e+01	2.113e+00	-5.822e-01	4.539e+00
ln(payroll)	1.207e+01	1.677e+00	3.700e-02	2.976e+00
ln(employment)	7.717e+00	1.770e+00	4.409e-02	3.008e+00
ln(age)	3.380e+00	4.555e-01	-2.243e+00	8.634e+00
ln(age)^2	1.163e+01	2.635e+00	-1.723e+00	5.351e+00

Panel B: Private Equity Held firm-Year Observations. N = 4,500*.

	mean	std dev.	skewness	kurtosis
asinh(research)	1.974e+00	3.335e+00	1.321e+00	3.224e+00
asinh(development)	3.696e+00	4.368e+00	5.210e-01	1.626e+00
ln(1+Engineers)	1.333e-01	6.845e-01	5.789e+00	3.833e+01
ln(1+Scientists)	4.490e-02	3.599e-01	9.323e+00	9.908e+01
ln(1+Technicians)	8.282e-01	1.456e+00	1.947e+00	6.687e+00
asinh(R&DCAPX)	2.220e+00	3.318e+00	1.092e+00	2.787e+00
Trade Secret Dummy	2.216e-01	4.153e-01	1.341e+00	2.798e+00
Trade Secret Missing	5.474e-01	4.978e-01	-1.904e-01	1.036e+00
Design Patents Important Dummy	3.656e-01	4.816e-01	5.582e-01	1.312e+00
Design Patents Important Missing	5.474e-01	4.978e-01	-1.904e-01	1.036e+00
Utility Patents Important Dummy	3.009e-01	4.587e-01	8.684e-01	1.754e+00

Utility Patents Important Missing	5.463e-01	4.979e-01	-1.859e-01	1.035e+00
ln(1+Patents Applied)	3.838e-01	9.567e-01	3.127e+00	1.439e+01
Patents Applied>0 Dummy	1.928e-01	3.946e-01	1.557e+00	3.424e+00
Missing Patent Applications	4.137e-01	4.926e-01	3.503e-01	1.123e+00
ln(1+Patents Issued)	2.734e-01	8.118e-01	4.004e+00	2.260e+01
Patents Issued > 0 Dummy	1.484e-01	3.556e-01	1.978e+00	4.911e+00
Patents Issued Missing	4.374e-01	4.961e-01	2.525e-01	1.064e+00
New Products Introduced Dummy	2.622e-01	4.399e-01	1.081e+00	2.169e+00
New Production Processes Dummy	1.279e-01	3.340e-01	2.228e+00	5.965e+00
New Logistics Processes Dummy	8.063e-02	2.723e-01	3.081e+00	1.049e+01
New Support Activities Dummy	1.378e-01	3.448e-01	2.101e+00	5.415e+00
ln(sales)	1.212e+01	1.709e+00	-2.636e-01	5.781e+00
ln(payroll)	1.068e+01	1.420e+00	7.673e-01	3.701e+00
ln(employment)	6.539e+00	1.492e+00	7.908e-01	3.934e+00
ln(age)	3.191e+00	6.338e-01	-1.651e+00	5.382e+00
ln(age)^2	1.059e+01	3.421e+00	-1.150e+00	3.260e+00

Panel C Other Private firms. N = 132,000*.

	mean	std dev.	skewness	kurtosis
asinh(research)	1.607e+00	2.880e+00	1.535e+00	4.022e+00
asinh(development)	2.781e+00	3.738e+00	8.056e-01	2.043e+00
ln(1+Engineers)	1.090e-01	5.562e-01	6.163e+00	4.702e+01
ln(1+Scientists)	3.452e-02	3.066e-01	1.138e+01	1.590e+02
ln(1+Technicians)	5.314e-01	1.052e+00	2.333e+00	8.786e+00
asinh(R&DCAPX)	1.588e+00	2.632e+00	1.399e+00	3.742e+00
Trade Secret Dummy	2.580e-01	4.375e-01	1.106e+00	2.224e+00
Trade Secret Missing	5.598e-01	4.964e-01	-2.410e-01	1.058e+00
Design Patents Important Dummy	3.724e-01	4.834e-01	5.279e-01	1.279e+00
Design Patents Important Missing	5.602e-01	4.964e-01	-2.426e-01	1.059e+00
Utility Patents Important Dummy	3.212e-01	4.670e-01	7.657e-01	1.586e+00
Utility Patents Important Missing	5.588e-01	4.965e-01	-2.370e-01	1.056e+00
ln(1+Patents Applied)	2.149e-01	6.847e-01	4.022e+00	2.182e+01
Patents Applied>0 Dummy	1.242e-01	3.298e-01	2.279e+00	6.194e+00
Missing Patent Applications	3.806e-01	4.855e-01	4.919e-01	1.242e+00
ln(1+Patents Issued)	1.452e-01	5.492e-01	4.975e+00	3.268e+01
Patents Issued > 0 Dummy	9.285e-02	2.902e-01	2.806e+00	8.873e+00
Patents Issued Missing	4.078e-01	4.914e-01	3.754e-01	1.141e+00
New Products Introduced Dummy	2.583e-01	4.377e-01	1.104e+00	2.220e+00
New Production Processes Dummy	1.497e-01	3.568e-01	1.964e+00	4.856e+00
New Logistics Processes Dummy	6.834e-02	2.523e-01	3.421e+00	1.271e+01
New Support Activities Dummy	1.315e-01	3.380e-01	2.181e+00	5.755e+00
ln(sales)	1.026e+01	1.859e+00	9.808e-02	5.673e+00
ln(payroll)	9.034e+00	1.375e+00	1.155e+00	4.650e+00
ln(employment)	4.857e+00	1.397e+00	9.422e-01	4.567e+00
ln(age)	2.950e+00	7.377e-01	-1.070e+00	3.452e+00
ln(age)^2	9.246e+00	3.821e+00	-5.837e-01	2.128e+00

* Note: The number of observations is rounded due to Census Bureau disclosure regulations.

Table 2

Linear Probability Models of Ownership Status

Parameter estimates and standard errors (in parentheses) for linear probability models of how various innovation strategy variables are related to equity ownership type. Panel A includes research and development variables; whereas, Panel B includes counts of scientists, engineers and technicians. In each specification, the dependent variable is an indicator taking a value of one or zero based on the distinction indicated at the head of each column. The sample in Column 1 includes all firm-year observations over 2009-2015 within the three ownership categories: Public, private-equity held and other private. Samples in Columns 2 and 3 contain the union of two subsamples indicated by the two categories in the heading. All regressions include six-digit NAICS industry-by-year fixed effects. Standard errors are computed by clustering on four-digit NAICS industry codes. The number of observations is rounded due to Census Bureau disclosure regulations. Statistical significance: 1% = ***, 5% = ** and 10% = *.

Panel A

	Public vs. All others	PE Held vs. Public	PE Held vs. Other Private
asinh(Research)	1.326e-03**	-4.994e-03***	-1.559e-03***
	(1.985e+00)	(-2.947e+00)	(-3.264e+00)
asinh(Development)	1.936e-03**	-0.0005823	1.036e-03**
	(2.557e+00)	(-2.925e-01)	(2.343e+00)
asinh(R&D CAPX)	2.699e-02***	-0.01109	-0.000661
	(4.073e+00)	(-6.820e-01)	(-1.517e-01)
Trade Secret Important Dummy	0.005334	0.0002060	-0.002588
	(6.260e-01)	(4.190e-03)	(-4.172e-01)
Trade Secret Importance Missing	-8.997e-03*	6.432e-02***	6.643e-03*
	(-1.710e+00)	(3.714e+00)	(1.890e+00)
Design Patents Important Dummy	-0.005904	8.593e-02*	0.002896
	(-5.354e-01)	(1.814e+00)	(5.457e-01)
Design Patent Importance Missing	1.820e-02***	0.005768	-0.003575
	(2.955e+00)	(5.604e-01)	(-1.150e+00)
Utility Patents Important Dummy	-1.466e-02*	-0.02321	1.585e-02***
	(-1.900e+00)	(-8.344e-01)	(3.041e+00)
Utility Patents Importance Missing	1.699e-02***	-0.04205	0.004453
	(3.877e+00)	(-1.440e+00)	(1.460e+00)
ln(1+ Patents Filed)	4.821e-02***	2.343e-02**	-0.00627
	(8.519e+00)	(2.254e+00)	(-1.528e+00)
Patents Filed >0 Dummy	-3.090e-02***	-1.060e-01***	0.007482
	(-3.412e+00)	(-4.327e+00)	(1.365e+00)
Patents Filed Missing	-9.067e-03**	0.02444	0.0004747
	(-2.535e+00)	(9.011e-01)	(1.666e-01)
ln(1+ Patents Issued)	-1.352e-02***	0.007850	-0.002666
	(-4.745e+00)	(5.733e-01)	(-1.296e+00)
Patents Issued>0 Dummy	-8.243e-03**	-3.258e-02*	-1.342e-02***
	(-2.325e+00)	(-1.763e+00)	(-4.404e+00)
Patents Issued Missing	1.899e-02***	-0.02369	0.003631
	(4.579e+00)	(-1.370e+00)	(1.378e+00)
New Products Introduced Dummy	-0.003485	0.02491	0.0003114
	(-1.187e+00)	(1.650e+00)	(1.447e-01)

New Production Processes Dummy	9.303e-03***	-1.680e-02***	2.320e-03***
	(7.766e+00)	(-2.889e+00)	(3.330e+00)
New Logistics Processes Dummy	4.591e-02***	-1.556e-01***	1.392e-02***
	(1.272e+01)	(-7.388e+00)	(6.507e+00)
New Support Activities Dummy	0.003943	1.049e-01***	1.353e-02***
	(1.345e+00)	(5.653e+00)	(6.996e+00)
ln(Sales)	0.008456	-2.341e-01***	-0.0009916
	(9.041e-01)	(-3.365e+00)	(-1.305e-01)
ln(payroll)	0.001483	2.782e-02**	0.0006968
	(7.398e-01)	(2.021e+00)	(4.368e-01)
ln(employees)	0.001074	0.001508	0.0001024
	(1.388e+00)	(7.191e-01)	(2.145e-01)
ln(age)	-0.002395	-0.01452	-0.003054
	(-7.067e-01)	(-8.770e-01)	(-1.199e+00)
ln(age)^2	0.01289	-0.01120	0.003656
	(1.540e+00)	(-2.492e-01)	(5.675e-01)
_cons	-5.443e-01***	2.017e+00***	-1.917e-01***
	(-1.722e+01)	(1.303e+01)	(-1.027e+01)
Adj R-squared	0.3174	0.3571	0.07389
Obs	148000	15500	137000

Panel B

	Public vs. All others	PE Held vs. Public	PE Held vs. Other Private
ln(1+Scientist Count)	1.526e-03**	-0.0004361	0.0002346
	(2.327e+00)	(-2.459e-01)	(5.661e-01)
ln(1+Engineer Count)	-0.001702	-0.01707	-0.003591
	(-5.054e-01)	(-1.046e+00)	(-1.429e+00)
ln(1+Technitian Count)	1.462e-02*	-0.006922	0.002821
	(1.718e+00)	(-1.532e-01)	(4.396e-01)
R&D CAPX	2.616e-02***	-0.01071	-0.0003525
	(4.012e+00)	(-6.675e-01)	(-8.231e-02)
Trade Secret Dummy	0.004537	-0.0008660	-0.002299
	(5.333e-01)	(-1.746e-02)	(-3.716e-01)
Trade Secret Missing	-8.642e-03*	6.699e-02***	6.658e-03*
	(-1.683e+00)	(3.832e+00)	(1.884e+00)
Design Patents Important Dummy	-0.005440	8.390e-02*	0.001972
	(-4.948e-01)	(1.750e+00)	(3.706e-01)
Design Patent Dummy Missing	1.671e-02***	0.004822	-0.003954
	(2.736e+00)	(4.834e-01)	(-1.277e+00)
Utility Patents Important Dummy	-0.01179	-0.02224	1.645e-02***
	(-1.506e+00)	(-8.203e-01)	(3.227e+00)
Utility Patents Dummy Missing	1.691e-02***	-0.04431	0.003869
	(3.935e+00)	(-1.550e+00)	(1.268e+00)
ln(1+# Patents Filed)	4.587e-02***	2.195e-02**	-0.006158
	(7.971e+00)	(2.128e+00)	(-1.567e+00)
Patents Filed >0 Dummy	-2.853e-02***	-1.039e-01***	0.007719
	(-3.107e+00)	(-4.341e+00)	(1.426e+00)
Patent Applications Missing	-9.099e-03**	0.02601	0.0008493
	(-2.580e+00)	(9.751e-01)	(3.012e-01)

ln(1+# Patents Issued)	-1.296e-02***	0.004444	-0.001868
	(-4.919e+00)	(3.301e-01)	(-9.317e-01)
Patents Issued>0 Dummy	-8.617e-03**	-3.865e-02**	-1.359e-02***
	(-2.459e+00)	(-2.109e+00)	(-4.604e+00)
Patents Issued Missing	1.872e-02***	-0.02255	0.003600
	(4.459e+00)	(-1.290e+00)	(1.373e+00)
New Products Introduced Dummy	-0.003912	2.859e-02*	0.0002537
	(-1.312e+00)	(1.914e+00)	(1.203e-01)
New Production Processes Dummy	9.186e-03***	-1.723e-02***	2.393e-03***
	(7.594e+00)	(-2.981e+00)	(3.493e+00)
New Logistics Processes Dummy	4.704e-02***	-1.611e-01***	1.408e-02***
	(1.249e+01)	(-7.619e+00)	(6.693e+00)
New Support Activities Dummy	0.001707	1.106e-01***	1.351e-02***
	(5.737e-01)	(5.998e+00)	(7.141e+00)
ln(Sales)	0.007406	-2.168e-01***	-0.0003666
	(7.838e-01)	(-3.140e+00)	(-4.818e-02)
ln(payroll)	0.001679	2.428e-02*	0.0005692
	(8.344e-01)	(1.775e+00)	(3.569e-01)
ln(employees)	4.971e-03*	0.00104	-0.002982
	(1.761e+00)	(2.177e-01)	(-1.472e+00)
ln(age)	1.115e-02***	-1.287e-02***	-0.001332
	(7.496e+00)	(-4.858e+00)	(-1.572e+00)
ln(age)^2	9.703e-03***	-0.001208	-0.0008422
	(4.462e+00)	(-2.842e-01)	(-4.544e-01)
_cons	-5.415e-01***	2.022e+00***	-1.927e-01***
	(-1.709e+01)	(1.307e+01)	(-1.023e+01)
Adj R-squared	0.3195	0.3569	0.07371
Obs	148000	15500	137000

Table 3

Efficiency of innovation

Results from regressions of innovation outcomes on contemporaneous and lagged transformed research and development. Also included are three lags of the dependent variable and additional controls. Panel A presents specifications without interactions. The specification for patents issued also includes three lags of patents applied, whose parameters are suppressed. Panels B and C present specifications with the same independent variables, but also include interactions between ownership status (public and private equity held) with contemporaneous development and research, respectively. For the sake of brevity, only the interaction terms are reported in Panels B & C. All specifications are estimated using OLS and standard errors are clustered by four-digit NAICS codes. 1%, 5% and 10% significance indicated by ***, ** and *, respectively.

Panel A

	ln(1+patents issued)	ln(1+patents applied)	New Product	New Production Process	New Logistics Process	New Support Service
asinh(Development)	4.662e-02***	6.252e-02***	3.798e-02***	1.748e-02***	9.419e-03***	1.525e-02***
	(1.831e+01)	(1.900e+01)	(2.275e+01)	(9.469e+00)	(1.265e+01)	(1.609e+01)
L1.asinh(Development)	-1.754e-02***	-2.273e-02***	-1.607e-02***	-8.013e-03***	-3.651e-03***	-6.198e-03***
	(-8.692e+00)	(-8.920e+00)	(-1.351e+01)	(-6.770e+00)	(-5.051e+00)	(-7.949e+00)
L2.asinh(Development)	-1.066e-02***	-9.945e-03***	-1.660e-03	-1.056e-03	-1.103e-03	-1.136e-03
	(-5.338e+00)	(-4.958e+00)	(-1.394e+00)	(-1.299e+00)	(-1.549e+00)	(-1.414e+00)
L3.asinh(Development)	-6.809e-03***	-7.732e-03***	-1.854e-03*	-1.094e-03	-4.931e-04	-1.440e-03**
	(-3.567e+00)	(-3.825e+00)	(-1.678e+00)	(-1.570e+00)	(-8.102e-01)	(-2.180e+00)
asinh(Research)	1.779e-02***	2.354e-02***	8.714e-03***	9.550e-03***	4.963e-03***	7.892e-03***
	(5.536e+00)	(4.772e+00)	(7.790e+00)	(8.912e+00)	(6.268e+00)	(7.348e+00)
L1.asinh(Research)	-7.165e-03***	-7.705e-03***	-6.570e-04	-2.600e-03**	-2.138e-03***	-2.186e-03**
	(-3.309e+00)	(-2.609e+00)	(-4.737e-01)	(-2.297e+00)	(-2.920e+00)	(-2.224e+00)
L2.asinh(Research)	1.525e-03	1.869e-03	5.288e-05	6.429e-04	3.988e-04	-4.081e-05
	(5.937e-01)	(7.174e-01)	(4.851e-02)	(7.369e-01)	(5.332e-01)	(-4.690e-02)
L3.asinh(Research)	-2.181e-03	-4.078e-03	-1.990e-03*	-9.399e-04	2.418e-04	-2.078e-04
	(-8.121e-01)	(-1.459e+00)	(-1.686e+00)	(-9.587e-01)	(3.284e-01)	(-2.881e-01)
Public	-1.902e-02	-1.663e-02	-1.259e-02	-2.031e-02**	-7.243e-03	-2.209e-02***
	(-9.931e-01)	(-7.117e-01)	(-1.268e+00)	(-2.310e+00)	(-1.036e+00)	(-2.704e+00)
PE Held	6.537e-02***	6.669e-02***	-1.747e-02**	-1.172e-02	2.229e-03	-1.381e-02*
	(3.194e+00)	(3.085e+00)	(-2.040e+00)	(-1.500e+00)	(3.612e-01)	(-1.809e+00)
L1.Y	3.038e-01***	4.326e-01***	3.730e-01***	4.130e-01***	4.108e-01***	3.845e-01***
	(1.340e+01)	(2.752e+01)	(3.804e+01)	(4.737e+01)	(3.672e+01)	(4.068e+01)
L2.Y	1.079e-01***	2.043e-01***	8.724e-02***	9.217e-02***	1.090e-01***	9.773e-02***
	(5.524e+00)	(1.444e+01)	(5.768e+00)	(8.251e+00)	(1.204e+01)	(9.831e+00)
L3.Y	1.146e-01***	1.447e-01***	9.419e-02***	9.456e-02***	7.355e-02***	6.864e-02***
	(8.252e+00)	(1.116e+01)	(1.159e+01)	(1.105e+01)	(7.419e+00)	(8.595e+00)
ln(payroll)	7.269e-02***	9.978e-02***	-3.580e-03	-1.325e-02**	9.905e-04	-2.209e-03
	(4.919e+00)	(5.992e+00)	(-5.216e-01)	(-2.562e+00)	(2.363e-01)	(-4.273e-01)
ln(employment)	-3.684e-02***	-4.995e-02***	6.737e-04	1.825e-02***	3.867e-03	6.692e-03
	(-2.904e+00)	(-3.245e+00)	(8.870e-02)	(3.409e+00)	(9.399e-01)	(1.305e+00)
ln(sales)	6.236e-03	7.733e-03	1.048e-03	-1.608e-03	2.034e-03*	3.192e-03
	(1.447e+00)	(1.630e+00)	(5.549e-01)	(-1.129e+00)	(1.789e+00)	(1.544e+00)
_cons	-6.468e-01***	-8.611e-01***	6.429e-02*	6.927e-02***	-4.958e-02**	-3.027e-02
	(-7.261e+00)	(-8.638e+00)	(1.908e+00)	(3.053e+00)	(-2.432e+00)	(-1.193e+00)
Adj R-squared	7.126e-01	7.018e-01	4.272e-01	3.862e-01	3.328e-01	3.055e-01
Obs	28000	28000	28000	28000	28000	28000

Panel B

	ln(1+patents issued)	ln(1+patents applied)	New Product	New Production Process	New Logistics Process	New Support Service
asinh(development)*PE_Held	1.156e-02**	1.017e-02	-4.127e-03	-2.532e-03	3.574e-04	-1.711e-03
	(2.275e+00)	(1.553e+00)	(-1.544e+00)	(-1.437e+00)	(2.179e-01)	(-9.752e-01)
asinh(development)*Public	4.043e-02***	4.048e-02***	-5.931e-03***	-2.273e-03	1.358e-03	-1.331e-03
	(9.763e+00)	(9.537e+00)	(-3.723e+00)	(-1.561e+00)	(1.261e+00)	(-1.093e+00)
Constant	-5.970e-01***	-8.010e-01***	5.023e-02	6.361e-02***	-4.627e-02**	-3.377e-02
	(-7.208e+00)	(-8.176e+00)	(1.441e+00)	(2.812e+00)	(-2.336e+00)	(-1.361e+00)
R-Squared	7.163e-01	7.048e-01	4.277e-01	3.863e-01	3.328e-01	3.055e-01
Observations	28000	28000	28000	28000	28000	28000

Panel C

	ln(1+patents issued)	ln(1+patents applied)	New Product	New Production Process	New Logistics Process	New Support Service
asinh(Research)*PE_Held	9.224e-03	1.120e-02	-5.332e-03	-3.443e-03	-1.524e-03	-2.021e-03
	(1.176e+00)	(1.250e+00)	(-1.391e+00)	(-1.346e+00)	(-6.805e-01)	(-7.736e-01)
asinh(Research)*Public	3.530e-02***	3.689e-02***	-4.933e-03***	-3.051e-03**	2.676e-04	-1.367e-03
	(7.997e+00)	(7.851e+00)	(-2.872e+00)	(-2.133e+00)	(1.875e-01)	(-7.936e-01)
Constant	-6.229e-01***	-8.272e-01***	5.639e-02	6.426e-02***	-4.932e-02**	-3.265e-02
	(-7.200e+00)	(-8.268e+00)	(1.612e+00)	(2.839e+00)	(-2.471e+00)	(-1.328e+00)
R-Squared	7.149e-01	7.039e-01	4.275e-01	3.863e-01	3.328e-01	3.055e-01
Observations	28000	28000	28000	28000	28000	28000

Table 4

Descriptive Statistics for Sample where Lead Minus Lag Changes can be Computed

The table below contains summary statistics for firm-year observations where at least one lead and lag of innovation input and output variables can be observed. All change variables are the lead minus the lag. N = 37500*.

	mean	Sd	skewness	kurtosis
pub_to_pe	8.319e-04	2.883e-02	3.463e+01	1.200e+03
pe_to_pub	6.977e-04	2.641e-02	3.782e+01	1.431e+03
pe_to_othpriv	2.335e-03	4.826e-02	2.062e+01	4.263e+02
othpriv_to_pe	5.018e-03	7.066e-02	1.401e+01	1.973e+02
othpriv_to_pub	1.878e-03	4.330e-02	2.301e+01	5.303e+02
chnglnsales	9.938e-02	1.036e+00	-1.110e-01	3.361e+01
chnglnpay	9.078e-02	4.228e-01	-4.760e+00	9.729e+01
chnglnemp	4.271e-02	4.049e-01	-3.795e+00	6.975e+01

*Note: The number of observations is rounded due to Census Bureau disclosure regulations.

Table 5

Effect of Changes of Ownership Status on Changes in Innovation Strategy.

Regressions of lead-minus-lag changes of innovation strategy variables on dummies indicating changes in equity ownership status. Research, development and R&D CAPX are all asinh transformed before changes are computed. Patents applied and patents issued are started-log transformed before changes are computed. Size control variables (sales, payroll and employment) are log transformed before changes are computed. Standard errors, in parentheses, are clustered by four-digit NAICS codes. Statistical significance is indicated by asterisks: *** is 1%, ** is 5%, and * is 10%, respectively.

	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8	Model 9
	Research	Development	R&D CAPX	Product Innovation	Process Innovation	Logistics Innovation	Support Innovation	Patents Applied	Patents Issued
pub_to_pe	0.08565	8.390e-01**	0.10360	0.04860	0.1075	0.07083	1.444e-01**	0.07039	-0.08703
	(2.023e-01)	(2.444e+00)	(2.161e-01)	(5.631e-01)	(1.494e+00)	(1.560e+00)	(2.318e+00)	(2.561e-01)	(-3.297e-01)
pe_to_pub	0.9449	0.2418	-0.5003	1.014e-02**	-1.059e-01*	-0.03334	-0.06554	0.1344	-0.02130
	(1.447e+00)	(4.403e-01)	(-8.110e-01)	(2.256e+00)	(-1.774e+00)	(-4.882e-01)	(-8.881e-01)	(8.795e-01)	(-1.414e-01)
pe_to_othpriv	0.3143	-0.04972	0.1662	0.01259	-9.331e-02**	-6.221e-02*	-0.05264	-0.06352	-0.1071
	(8.262e-01)	(-1.157e-01)	(3.959e-01)	(2.542e-01)	(-2.216e+00)	(-1.803e+00)	(-1.166e+00)	(-5.437e-01)	(-1.107e+00)
othpriv_to_pe	0.2355	-0.1675	0.2691	0.02780	-0.02750	0.005815	-0.01368	0.07542	0.06682
	(8.111e-01)	(-5.225e-01)	(9.417e-01)	(7.417e-01)	(-1.023e+00)	(2.397e-01)	(-4.830e-01)	(1.416e+00)	(1.486e+00)
othpriv_to_pub	-0.4500	-0.1477	-0.6087	0.06154	9.385e-02**	-0.01000	0.009711	0.07559	-0.05334
	(-9.320e-01)	(-3.207e-01)	(-1.233e+00)	(1.102e+00)	(2.530e+00)	(-2.323e-01)	(2.167e-01)	(6.667e-01)	(-3.960e-01)
chnglnsales	1.026e-01***	1.226e-01***	1.293e-01***	4.875e-03*	-0.0004877	-0.001724	0.001005	2.038e-02***	1.116e-02**
	(5.306e+00)	(3.795e+00)	(5.633e+00)	(1.820e+00)	(-2.578e-01)	(-9.488e-01)	(4.911e-01)	(4.539e+00)	(2.208e+00)
chnglnpay	2.623e-01***	5.086e-01***	3.884e-01***	4.903e-02***	1.586e-02*	1.170e-02*	0.006848	9.567e-02***	5.851e-02***
	(3.444e+00)	(6.678e+00)	(4.176e+00)	(5.299e+00)	(1.866e+00)	(1.749e+00)	(7.957e-01)	(4.533e+00)	(3.398e+00)
chnglnemp	-0.02130	-1.759e-01*	-1.557e-01*	-3.974e-02***	-0.01356	-0.007685	0.004987	-0.008789	-0.004689
	(-2.260e-01)	(-1.930e+00)	(-1.872e+00)	(-3.452e+00)	(-1.222e+00)	(-9.907e-01)	(5.245e-01)	(-4.618e-01)	(-2.597e-01)
_cons	-5.505e-02***	-1.150e-01***	0.02041	-1.428e-02***	-1.066e-02***	-6.394e-03***	-1.382e-02***	-4.379e-02***	-1.325e-02***
	(-2.972e+00)	(-4.912e+00)	(6.443e-01)	(-4.017e+00)	(-3.715e+00)	(-3.502e+00)	(-5.158e+00)	(-4.894e+00)	(-2.806e+00)
Adj R-squared	0.002140	0.003507	0.003136	0.0005144	0.0001639	-0.000004443	0.00007243	0.002638	0.001210
Obs	37500	37500	37500	37500	37500	37500	37500	37500	37500

Table 6
Exit Outcomes of Private Equity Investments

This table presents parameter estimates (and standard errors in parentheses) of linear probability models of exit outcome type for a private equity portfolio company. Five equestions are jointly estimated for five exit outcomes (trade sale, IPO, secondary buyout, bankruptcy, or still held as of 2015), but only the first three are reported due to disclosure limitations. Independent variables include various measures of the firm's innovation strategy averaged over all the years, for which we have data, where the firm was part of the private equity portfolio. Industry-by-year fixed effects are included in all regressions, where industry is defined by 3-digit NAICS code. Standard errors are clustered by 3-digit industry. Statistical significance at the 1%, 5% and 10% levels are indicated by ***, ** and *.

	Trade Sale	IPO	Secondary Buyout
asinh(research)	8.317e-03*	-4.403e-03	-1.189e-03
	(1.855e+00)	(-1.482e+00)	(-3.390e-01)
asinh(development)	-4.843e-03	-8.669e-04	4.457e-03
	(-7.710e-01)	(-2.956e-01)	(7.671e-01)
asinh(R&D CAPX)	8.797e-04	3.523e-03	-6.436e-03
	(1.256e-01)	(9.772e-01)	(-1.007e+00)
New Products or Services	9.327e-02**	-7.794e-03	-6.001e-02*
	(2.219e+00)	(-3.759e-01)	(-1.664e+00)
New Production Processes	4.413e-02	-9.012e-03	-3.438e-02
	(9.548e-01)	(-4.372e-01)	(-8.034e-01)
New Logistics Processes	-1.734e-02		1.575e-02
	(-2.854e-01)		(3.176e-01)
New Support Services	-4.122e-02	1.343e-02	3.914e-02
	(-7.457e-01)	(4.821e-01)	(8.938e-01)
Trade Secrets Dummy	1.060e-01***	-7.380e-03	-8.574e-02**
	(3.013e+00)	(-3.530e-01)	(-2.053e+00)
Trade Secrets Missing	3.209e-01**	-5.900e-02	-3.897e-01***
	(2.435e+00)	(-9.912e-01)	(-5.732e+00)
Design Patents Important	9.508e-02	7.311e-03	-1.477e-01***
	(1.502e+00)	(2.689e-01)	(-3.754e+00)
Design Patents Missing	-1.583e-01	-5.427e-02	2.450e-01
	(-3.980e-01)	(-1.038e+00)	(6.764e-01)
Utility Patents Important	-1.375e-01***	-3.172e-02	1.935e-01***
	(-3.058e+00)	(-1.283e+00)	(4.338e+00)
Utility Patents Missing	-1.582e-01	8.722e-02	1.584e-01
	(-3.759e-01)	(1.139e+00)	(4.197e-01)
ln(employment)	-5.137e-02***	-1.450e-02	4.886e-02***
	(-3.108e+00)	(-1.435e+00)	(3.063e+00)
ln(payroll)	1.133e-02	5.172e-02***	-5.086e-02**
	(6.066e-01)	(4.383e+00)	(-2.495e+00)
ln(age)	-4.956e-02	-4.831e-02	1.634e-01*
	(-5.496e-01)	(-1.258e+00)	(1.811e+00)
ln(age)^2	7.124e-03	6.614e-03	-2.769e-02
	(4.173e-01)	(9.101e-01)	(-1.610e+00)
asinh(patents applied)	3.219e-02*	5.097e-03	-6.838e-02***
	(1.792e+00)	(3.154e-01)	(-4.720e+00)
Patents Applied Dummy	-6.888e-02	-1.235e-02	X

	(-1.485e+00)	(-4.321e-01)	X
ln(years held)	8.787e-02**	-4.743e-02*	-3.043e-02
	(2.114e+00)	(-1.913e+00)	(-6.860e-01)
syndicated	-2.281e-02	3.756e-02**	-1.877e-02
	(-1.216e+00)	(2.470e+00)	(-8.111e-01)
R-Squared	0.1071	0.1755	0.1147
N	1800	1800	1800